

Shayan Zargari

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Professional Summary

Wireless Systems Engineer and Ph.D. in Electrical Engineering specializing in Wi-Fi sensing, mmWave radar, signal processing, optimization, and AI/ML-enabled wireless systems. Experienced in designing, implementing, and evaluating advanced wireless algorithms for Wi-Fi systems. Strong publication record in IEEE venues with hands-on expertise in real-world wireless performance evaluation.

Technical Skills

Programming: Python, MATLAB, C/C++, Bash, Git/GitHub, Linux

Machine Learning: TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, SciPy, Pandas, CNNs, LSTMs, GANs, Autoencoders, Transfer Learning, OOD Detection

Wireless Systems: Wi-Fi, IEEE 802.11, 5G NR, LTE, mmWave Radar, Wi-Fi Sensing, CSI Processing, MIMO, OFDM, Beamforming, Channel Estimation, Signal Detection

Optimization: Convex Optimization, CVX, SDR, Resource Allocation, Alternating Optimization

Educational Background

University of Alberta

Ph.D., Electrical Engineering (Wireless Communication Systems)

GPA: 4.0/4.0

Edmonton, Canada

May 2021–Apr 2025

Iran University of Science and Technology

M.Sc., Electrical Engineering (Wireless Communication Systems)

GPA: 4.0/4.0

Tehran, Iran

Aug 2018–Dec 2020

Islamic Azad University South Tehran Branch

B.Sc., Electrical Engineering (Electronics)

GPA: 4.0/4.0

Tehran, Iran

Aug 2014–Aug 2018

Research Experience

Wireless Communication Laboratory, University of Alberta

Researcher

May 2021–Apr 2025

- Lead researcher in developing 6G communication systems using signal processing and machine learning techniques.
- Intelligent reflecting surfaces (IRS), unmanned aerial vehicles (UAV), simultaneous wireless information and power transfer (SWIPT), and ambient backscatter communication (AmBC).

Wireless Communication Laboratory, Iran University of Science and Technology

Research Assistant

Aug 2018–Dec 2020

- Designed advanced optimization algorithms for resource allocation in 5G/6G systems.
- Focused on improving IRS-aided SWIPT systems, enhancing power and data transfer efficiency.

Optical Wireless Networks Laboratory, Iran University of Science and Technology

Research Assistant

Aug 2019–Aug 2020

- Developed resource allocation strategies for visible light communication (VLC) networks.
- Optimized data rates to establish high-speed and reliable VLC systems.

Wireless Communication Laboratory, Tarbiat Modares University

Visiting Researcher

Aug 2020–Aug 2021

- Optimized resource allocation in UAV-aided IRS systems using reinforcement learning techniques.
- Implemented reinforcement learning to predict and improve network performance dynamically.

Wireless Communication Laboratory, Sharif University of Technology

Visiting Researcher

Aug 2019–Mar 2020

- Designed device-to-device (D2D) communication systems using non-orthogonal multiple access (NOMA).
- Enhanced spectral efficiency by optimizing network models to reduce interference.

Honors & Awards

2023–2025: Recipient, Alberta Innovates Graduate Student Scholarship, University of Alberta

2024: Recipient, Variperm Mousavi-Daneshmand Engineering Innovation Award, University of Alberta

2022: Recipient, Alberta Graduate Excellence Scholarship, University of Alberta

2022: Ranked 1st among 39 M.Sc. students at Iran University of Science and Technology

2021: Recognized as an exceptional talent for graduate studies

2020: Top 5% in M.Sc. Entrance Exam among 10,555 candidates

Teaching Experience

University of Alberta

Teaching Assistant, Introduction to Communication Systems

Sep–Dec 2024

- Led lab sessions focused on the basics of analog/digital communication

University of Alberta

Teaching Assistant, Computer Programming for Engineers

Sep–Dec 2021 / Sep–Apr 2023

- Supported students in learning Python and C++ programming.

University of Alberta

Teaching Assistant, Analog Electronics

Jan–Apr (2021/2022)

- Guided students through hands-on labs in analog circuit design and testing.

University of Alberta

Teaching Assistant, Discrete Time Signals and Systems

Sep– Dec 2022

- Led lab sessions focused on discrete-time system analysis and digital filtering.

Iran University of Science and Technology

Teaching Assistant, Cellular Wireless Communications

Sep–Dec 2019

- Assisted in teaching cellular network technologies, including 4G/LTE and 5G systems.

Professional Experience

Huawei Technologies Canada

Ottawa, Canada

Engineer

Feb 2025–Present

- Develop and evaluate advanced Wi-Fi algorithms for sensing, radar-based perception, and wireless system optimization.
- Implement signal processing and machine learning pipelines for gesture recognition, activity sensing, and wireless performance analysis.
- Design experimental and simulation-based evaluation frameworks to assess algorithm robustness under realistic deployment conditions.
- Analyze CSI/radar/sensing data to extract discriminative features and improve detection, classification, and false-alarm performance.
- Support R&D activities related to Wi-Fi and mmWave radar sensing systems.
- Collaborate with cross-functional research and engineering teams to translate algorithmic concepts into practical system-level solutions.

University of Alberta

Edmonton, Canada

Collaborator with Huawei

May 2021–Jun 2023

- Collaborated with Huawei on cutting-edge research projects on 6G communication systems.
- Regularly coordinated with Huawei engineers to align research objectives with industrial goals.
- Presented research progress to Huawei's R&D team, contributing to developing new wireless technologies.

- Collaborated with the R&D team on developing video intercom systems.
- Solved complex circuit design problems and improved technical system efficiency.

Leading Journals and Global Conferences

Peer Reviewer

Aug 2020–Present

- Reviewed manuscripts for top-tier journals such as IEEE Access, IEEE Sens. J., IEEE IoT J., IEEE Commun. Lett., IEEE Wireless Commun. Lett., IEEE Trans. Commun., IEEE Trans. Veh. Technol., IEEE Trans. Wireless Commun., IEEE Trans. Signal Process., and IEEE J. Sel. Areas Commun. IEEE Veh. Technol. Mag.
- Evaluated papers for leading conferences, including:
 - 2020–2022 IEEE GLOBECOM (Wireless Commun., Commun. Theory)
 - 2022 IEEE/CIC ICC (Wireless Commun.)
 - 2023 IEEE GLOBECOM (RIS-Empowered 3CLS for 6G & Beyond Workshop)
 - 2024 IEEE JSAC (Next-Gen. Adv. Transceiver Tech. Spec. Issue)
 - 2025 IEEE WCNC (Wireless Commun. & Netw. Conf.)

Publications

- [1] **S. Zargari**, A. Khalili, and R. Zhang, "Energy efficiency maximization via joint active and passive beamforming design for multiuser MISO IRS-aided SWIPT," *IEEE Wirel. Commun. Lett.*, vol. 10, no. 3, pp. 557–561, 2021.
- [2] **S. Zargari**, A. Khalili, Q. Wu, M. R. Mili, and D. W. K. Ng, "Max-min fair energy-efficient beamforming design for intelligent reflecting surface-aided SWIPT systems with non-linear energy harvesting model," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5848–5864, 2021.
- [3] A. Khalili, **S. Zargari**, Q. Wu, D. W. K. Ng, and R. Zhang, "Multi-objective resource allocation for IRS-aided SWIPT," *IEEE Wirel. Commun. Lett.*, vol. 10, no. 6, pp. 1324–1328, 2021.
- [4] A. Khalili, E. M. Monfared, **S. Zargari**, M. R. Javan, N. M. Yamchi, and E. A. Jorswieck, "Resource management for transmit power minimization in UAV-assisted RIS HetNets supported by dual connectivity," *IEEE Trans. Wirel. Commun.*, vol. 21, no. 3, pp. 1806–1822, 2022.
- [5] **S. Zargari**, S. Farahmand, B. Abolhassani, and C. Tellambura, "Robust active and passive beamformer design for IRS-aided downlink MISO PS-SWIPT with a nonlinear energy harvesting model," *IEEE Trans. Green Commun. Netw.*, vol. 5, no. 4, pp. 2027–2041, 2021.
- [6] S. Bayat, A. Khalili, **S. Zargari**, M. R. Mili, and Z. Han, "Multi-objective resource allocation for D2D and enabled MC-NOMA networks by tchebycheff method," *IEEE Trans. Veh. Technol.*, vol. 70, no. 5, pp. 4464–4470, 2021.
- [7] **S. Zargari**, M. Kolivand, S. A. Nezamalhosseni, B. Abolhassani, L. R. Chen, and M. H. Kahaei, "Resource allocation of hybrid VLC/RF systems with light energy harvesting," *IEEE Trans. Green Commun. Netw.*, vol. 6, no. 1, pp. 600–612, 2022.
- [8] A. Hakimi, **S. Zargari**, C. Tellambura, and S. Herath, "IRS-enabled backscattering in a downlink non-orthogonal multiple access system," *IEEE Commun. Lett.*, vol. 26, no. 12, pp. 2984–2988, 2022.
- [9] **S. Zargari**, A. Hakimi, C. Tellambura, and S. Herath, "Multiuser MISO PS-SWIPT systems: Active or passive RIS?" *IEEE Commun. Lett.*, vol. 11, no. 9, pp. 1920–1924, 2022.
- [10] A. Hakimi, **S. Zargari**, C. Tellambura, and S. P. Herath, "Sum rate maximization of full-duplex MIMO monostatic backscatter networks under residual self-interference," in *17th Canadian Workshop on Information Theory, CWIT 2022, Ottawa, ON, Canada, June 5-8, 2022*. IEEE, 2022, pp. 103–108.
- [11] **S. Zargari**, C. Tellambura, and S. Herath, "Energy-efficient hybrid offloading for backscatter-assisted wirelessly powered MEC with reconfigurable intelligent surfaces," *IEEE Trans. Mob. Comput.*, vol. 22, no. 9, pp. 5262–5279, 2023.
- [12] **S. Zargari**, S. Farahmand, and B. Abolhassani, "Joint design of transmit beamforming, IRS platform, and power splitting SWIPT receivers for downlink cellular multiuser MISO," *Phys. Commun.*, vol. 48, p. 101413, 2021.

- [13] **S. Zargari**, A. Hakimi, C. Tellambura, and S. Herath, "User scheduling and trajectory optimization for energy-efficient IRS-UAV networks with SWIPT," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 1815–1830, 2023.
- [14] **S. Zargari**, H. Ahmadinejad, B. Abolhassani, and A. Falahati, "SWIPT-NOMA in cell-free massive MIMO," in *2020 28th Iranian Conference on Electrical Engineering (ICEE)*, 2020, pp. 1–6.
- [15] A. Hakimi, **S. Zargari**, C. Tellambura, and S. Herath, "Sum rate maximization of MIMO monostatic backscatter networks by suppressing residual self-interference," *IEEE Trans. Commun.*, vol. 71, no. 1, pp. 512–526, 2023.
- [16] **S. Zargari**, C. Tellambura, and A. Maaref, "Improved signal detection for ambient backscatter communications," *IEEE Trans. Veh. Technol.*, Jun. 2023.
- [17] A. Rezaei, A. Falahati, and **S. Zargari**, "Energy efficient multi-way massive MIMO AF relaying with imperfect CSI and optimal power allocation," *Wirel. Pers. Commun.*, vol. 132, pp. 1681–1702, 2023.
- [18] **S. Zargari**, A. Hakimi, C. Tellambura, and A. Maaref, "Enhancing AmBC systems with deep learning for joint channel estimation and signal detection," *IEEE Trans. Commun.*, pp. 1–1, 2023.
- [19] **S. Zargari**, A. Hakimi, F. Rezaei, C. Tellambura, and A. Maaref, "Signal detection in ambient backscatter systems: Fundamentals, methods, and trends," *IEEE Access*, vol. 11, pp. 140 287–140 324, 2023.
- [20] D. Galappaththige, **S. Zargari**, C. Tellambura, and G. Y. Li, "Near-field ISAC: Beamforming for multi-target detection," *IEEE Wirel. Commun. Lett.*, pp. 1–1, 2024.
- [21] **S. Zargari**, C. Tellambura, A. Maaref, and G. Y. Li, "Deep conditional generative adversarial networks for efficient channel estimation in AmBC systems," *Accepted, IEEE Trans. Mach. Learn. Commun. Netw.*, Jun. 2024.
- [22] M. Yousefpoori-Naeim, **S. Zargari**, and Z. Hatami, "Using machine learning to predict item difficulty and response time in medical tests," in *Proc. 19th Workshop Innov. Use of NLP for Build. Educ. Appl. (BEA 2024)*. Mexico City, Mexico: Association for Computational Linguistics, Jun. 2024, pp. 551–560.
- [23] A. Hakimi, **S. Zargari**, and C. Tellambura, "Integrated communication and sensing with symbiotic radio and RIS-assisted backscatter," *IEEE Wireless Commun. Lett.*, pp. 1–1, 2024.
- [24] **S. Zargari**, D. Galappaththige, C. Tellambura, and H. Vincent Poor, "A Riemannian manifold approach to constrained resource allocation in ISAC," *IEEE Trans. Commun.*, pp. 1–1, 2024.
- [25] **S. Zargari**, D. Galappaththige, and C. Tellambura, "Transmit power-efficient beamforming design for integrated sensing and backscatter communication," *IEEE open j. Commun. Soc.*, pp. 1–1, 2024.
- [26] D. Galappaththige, **S. Zargari**, C. Tellambura, and G. Y. Li, "Optimization of rate-splitting multiple access with integrated sensing and backscatter communication," *IEEE Trans. Veh. Technol.*, vol. 74, pp. 9117–9133, 2025.
- [27] —, "Low-complexity multi-target detection in ELAA ISAC," *IEEE Commun. Lett.*, 2025.
- [28] **S. Zargari**, D. Galappaththige, C. Tellambura, and G. Y. Li, "Downlink beamforming for cell-free ISAC: A fast complex oblique manifold approach," *IEEE Trans. Wirel. Commun.*, 2025.
- [29] D. Galappaththige, **S. Zargari**, and C. Tellambura, "Dual function of sensing and backscatter communication in cellular networks," *IEEE Internet Things J.*, vol. 8, pp. 64–71, 2025.
- [30] **S. Zargari**, D. Galappaththige, and C. Tellambura, "Low-complexity CRB minimization for ISAC with a generalized target response matrix," *IEEE Wirel. Commun. Lett.*, 2025.